IT Workshop II (Cloud Computing): MapReduce - Concepts

September 20, 2026

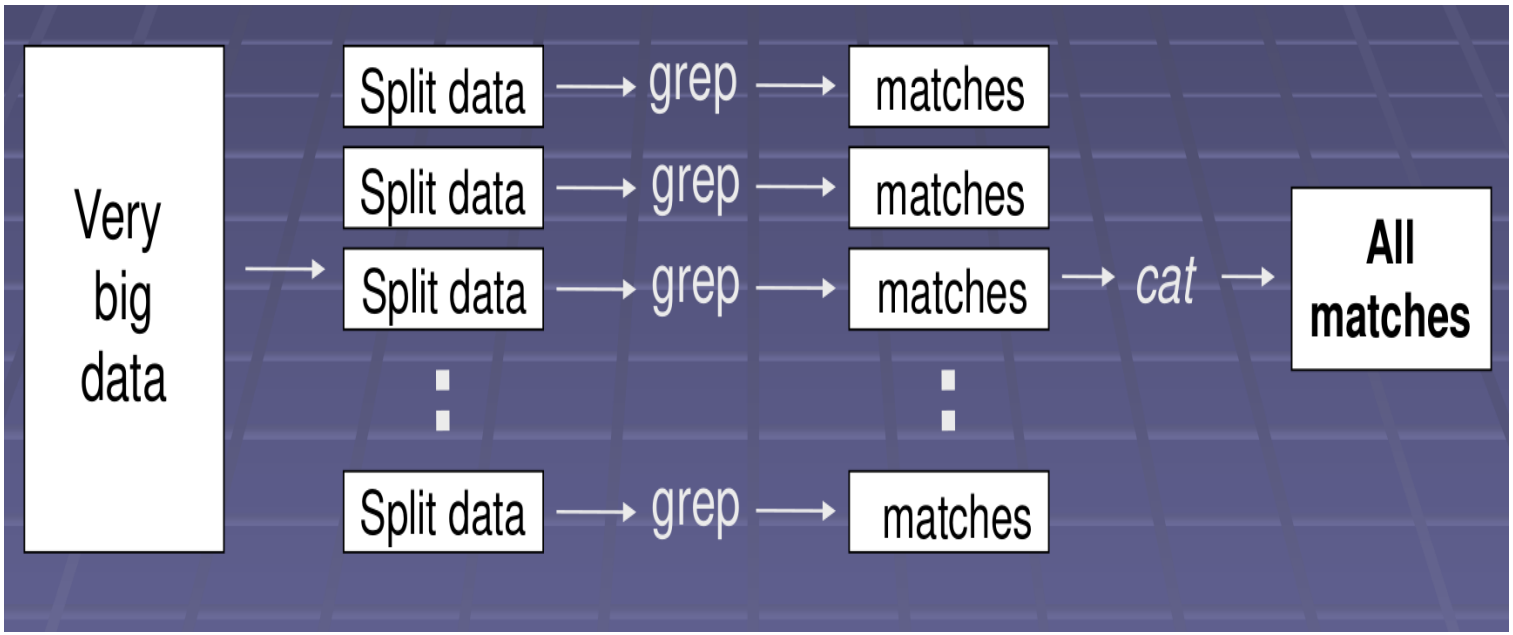
What is MapReduce?

- MapReduce is a simple programming model.
- MapReduce is useful for large-scale data processing. It distributes the processing of data.
- MapReduce offers high availability.
- The framework is essential for scalable data processing and has been widely used in big data environments, like Hadoop.
- The MapReduce process consists of two primary functions: Map Function and Reduce Function.

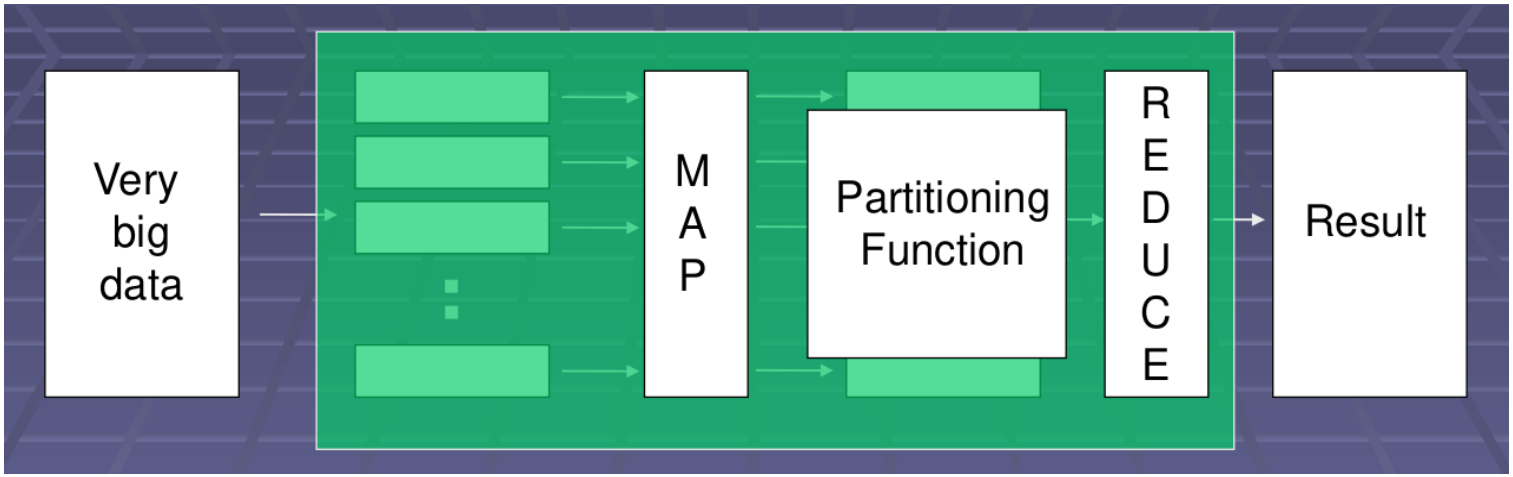
Motivation

- There is a demand for very large scale data processing.
- Lots of systems are required for processing (scaling).
- **Map Function:** Takes a set of data and converts it into key-value pairs. This phase processes each input independently, and the output is a collection of intermediate key-value pairs.
- **Reduce Function:** After the map phase, all the intermediate key-value pairs with the same key are grouped together. The reduce function processes these grouped values to aggregate the result (e.g., summing up counts of the same word).

Distributed Search

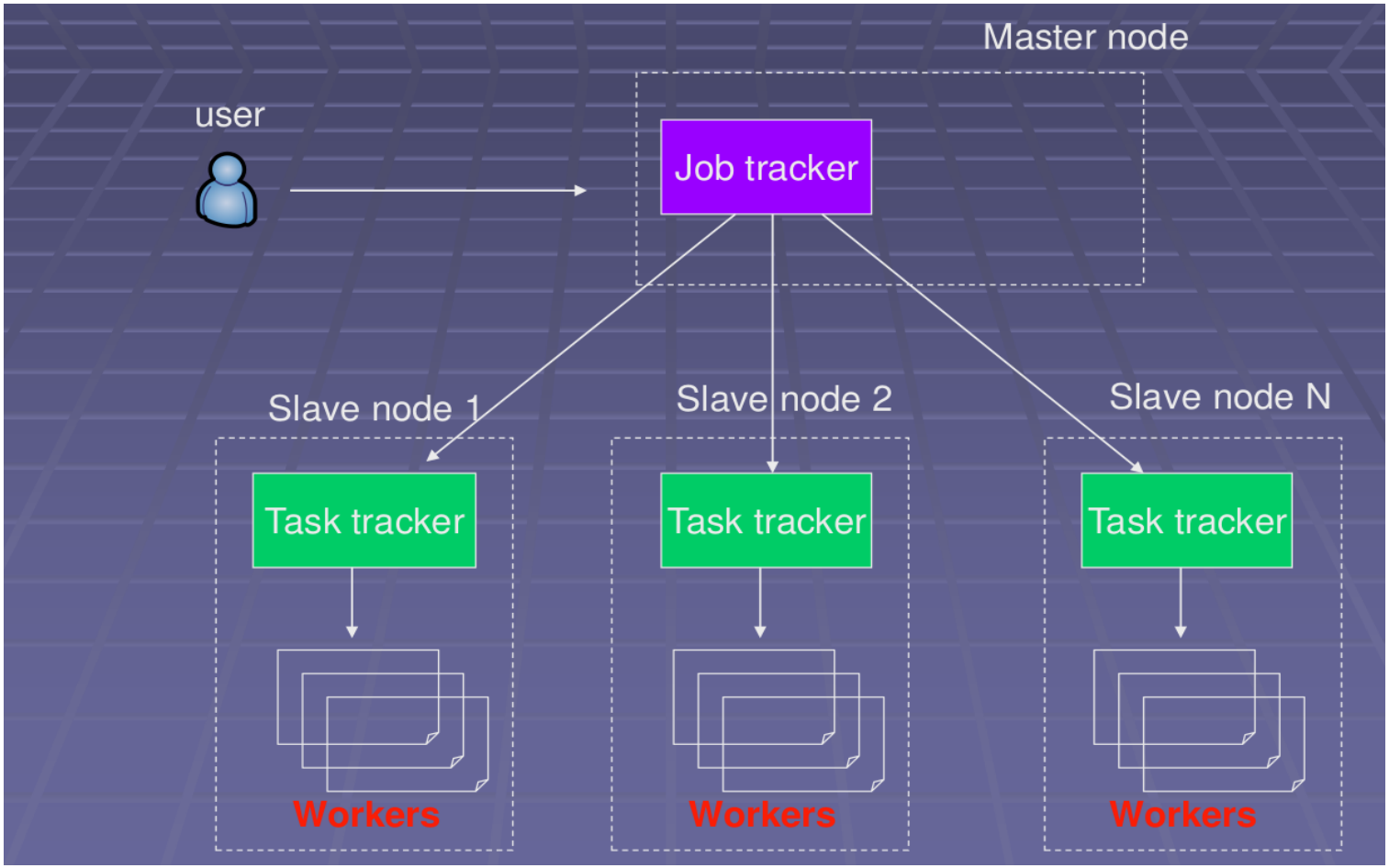


MapReduce

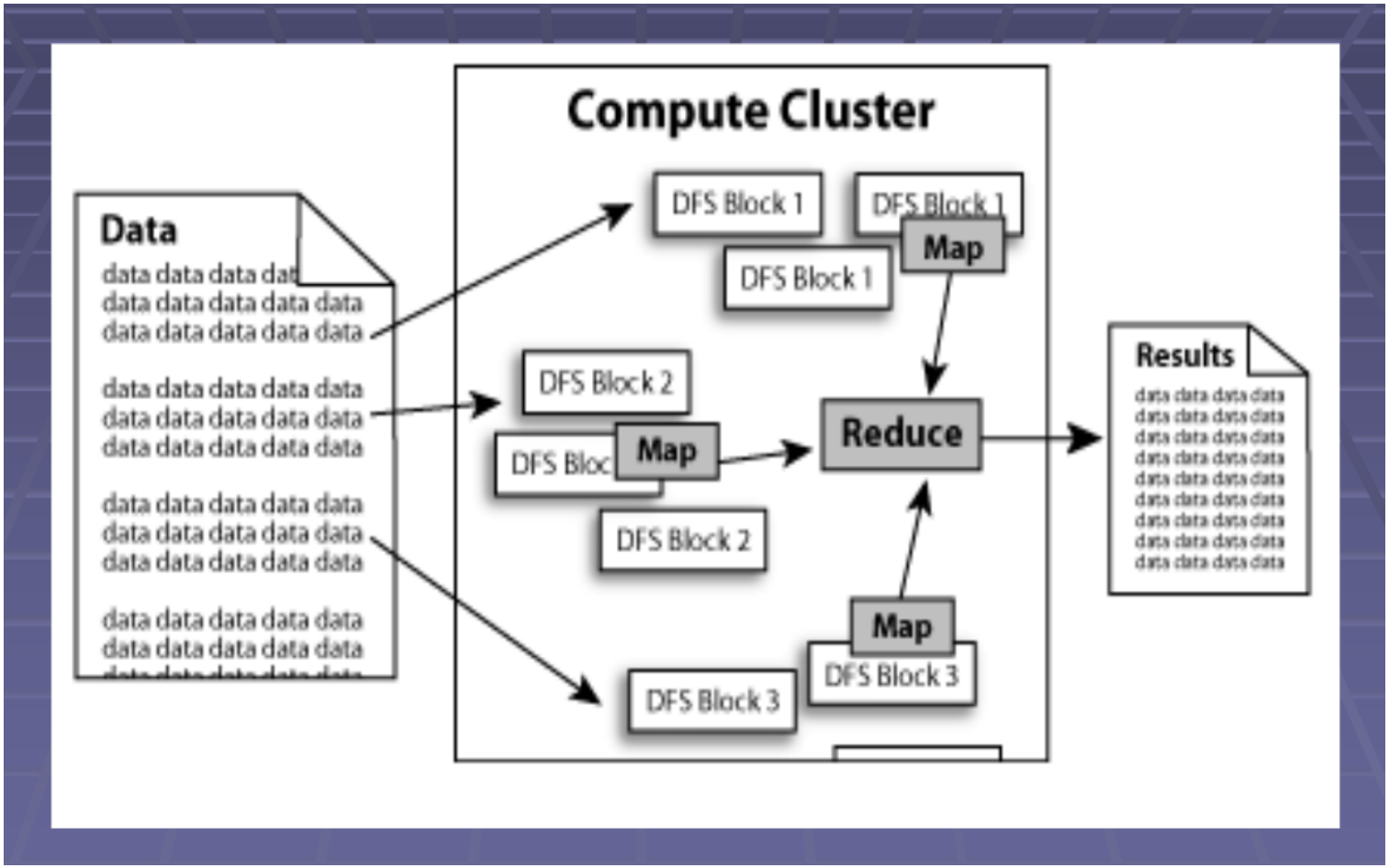


- **Partitioning Function:** A partitioner works like a condition in processing an input dataset. The partition phase takes place after the Map phase and before the Reduce phase. The number of partitioners is equal to the number of reducers. That means a partitioner will divide the data according to the number of reducers.

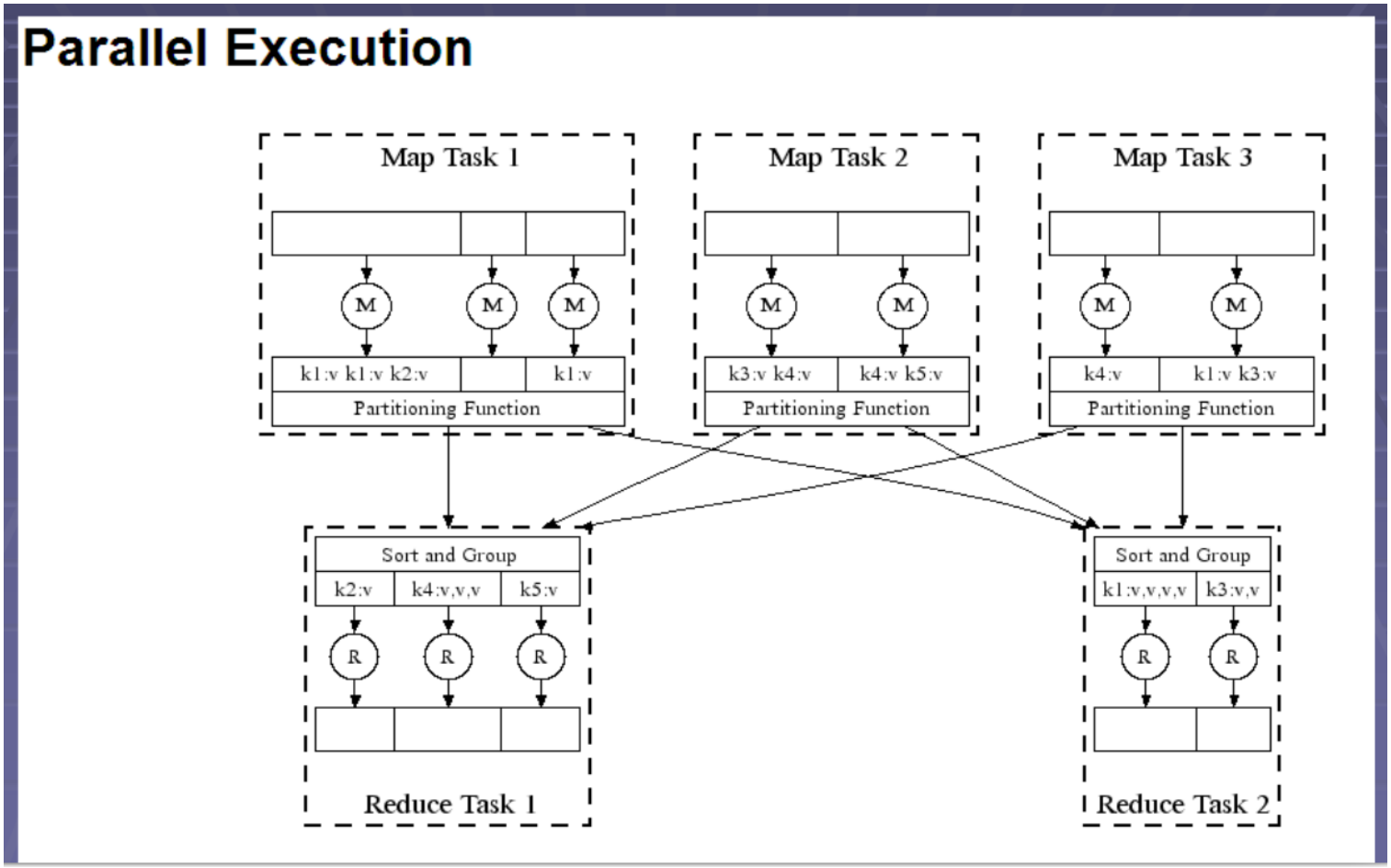
Architecture Overview



MapReduce


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DFS Block 1
DFS Block 1
Map
DFS Block 1
DFS Block 2
DFS Bloc
Map
Reduce
DFS Block 2
Map
DFS Block 3
DFS Block 3
Results
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MapReduce


Parallel Execution
Map Task 1
Map Task 2
Map Task 3
M
k1:v k1:v k2:v
k1:v
k3:v k4:v
k4:v k5:v
k4:v
k1:v k3:v
Partitioning Function
Sort and Group
k2:v
k4:v,v,v
k5:v
k1:v,v,v,v
k3:v,v
R
Reduce Task 1
Reduce Task 2

Fault Tolerance

- **Worker failure:** Workers are periodically pinged by master. If the processor of a worker fails, the tasks of that worker are reassigned to another worker.
- **Master failure:** Master writes periodic checkpoints. Another master can be started from the last checkpointed state. If eventually the master dies, the job will be aborted.
- Whenever either the primary or the backup executions finishes, mark it as completed.
- **Input error: bad records:** Map/Reduce functions sometimes fail for particular inputs. If master sees two failures for same record, next worker is told to skip the record.

MapReduce : Example

If you want to count the frequency of words in multiple documents:

- **Map Step:** A document is split into words. For each word, a key-value pair is created, where the key is the word and the value is 1.
- **Shuffle and Sort:** All the values are shuffled so that instances of the same word are grouped together.
- **Reduce Step:** The reduce function aggregates the values by summing them, giving the total count for each word.

The End